

Bangladesh University of Business and Technology



LAB REPORT

Experiment No : 01
Course Title : Operating System Lab
Course Code : CSE 210
Experiment Name: CPU Scheduling Algorithms

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Question 1. Write a C++ program for FCFS (First Come First Serve) CPU scheduling. Take the Number of Processes, Arrival Time (AT) and Burst Time (BT) as input. Calculate Completion Time (CT), Turnaround Time (TAT), and Waiting Time (WT).

Steps To Solve:

- Take the number of processes as input.
- Enter the Arrival Time (AT) and Burst Time (BT) for each process.
- Arrange the processes in ascending order according to the Arrival Time.
- Calculate the Completion Time (CT) for each process
 - For the first process: $CT = AT + BT$
 - For the next process: $CT = \text{Previous CT} + BT$.
- Calculate the Turnaround Time (TAT)
 - $TAT = CT - AT$
- Calculate the Waiting Time (WT)
 - $WT = TAT - BT$
- Display the Process ID, Arrival Time, Burst Time, Completion Time, Turnaround Time and Waiting Time.

Code:

```
#include <iostream>
using namespace std;
int main()
{
    int n;
    cout << "Enter number of processes: ";
    cin >> n;
    int A[n], B[n], C[n], TA[n], WT[n];
    for (int i = 0; i < n; i++)
    {
        cout << "Enter Arrival Time and Burst Time for Process "
              << i + 1 << ": ";
        cin >> A[i] >> B[i];
    }
```

```

for (int i = 0; i < n; i++)
{
    for (int j = i + 1; j < n; j++)
    {
        if (A[i] > A[j])
        {
            swap(A[i], A[j]);
            swap(B[i], B[j]);
        }
    }
}
C[0] = A[0] + B[0];
for (int i = 1; i < n; i++)
{
    if (C[i - 1] < A[i])
        C[i] = A[i] + B[i];
    else
        C[i] = C[i - 1] + B[i];
}
for (int i = 0; i < n; i++)
{
    TA[i] = C[i] - A[i];
    WT[i] = TA[i] - B[i];
}
cout << "\nAT\tBT\tCT\tTA\tWT\n";
for (int i = 0; i < n; i++)
{
    cout << A[i] << "\t"
        << B[i] << "\t"
        << C[i] << "\t"
        << TA[i] << "\t"
        << WT[i] << endl;
}
return 0;
}

```

Output:

```
PS E:\#\S-05-CSE-208\Lab Report 1> ./FIFO.exe
Enter number of processes: 5
Enter Arrival Time and Burst Time for Process 1: 1 8
Enter Arrival Time and Burst Time for Process 2: 3 7
Enter Arrival Time and Burst Time for Process 3: 2 9
Enter Arrival Time and Burst Time for Process 4: 4 3
Enter Arrival Time and Burst Time for Process 5: 2 3
```

AT	BT	CT	TA	WT
1	8	9	8	0
2	9	18	16	7
2	3	21	19	16
3	7	28	25	18
4	3	31	27	24

Question 2. Write a C++ program for SJF (Shortest Job First) CPU scheduling. Take the Number of Processes, Arrival Time (AT) and Burst Time (BT) as input. Calculate Completion Time (CT), Turnaround Time (TAT), and Waiting Time (WT).

Steps To Solve:

- Take the number of processes as input.
- Enter the Arrival Time (AT) and Burst Time (BT) for each process.
- Check which processes have arrived and select the process with the shortest Burst Time.
- Execute the selected process completely and calculate its Completion Time (CT).
- Repeat the process until all processes are completed.
- Calculate the Turnaround Time (WT)
 - $TAT = CT - AT$
- Calculate the Waiting Time (WT)
 - $WT = TAT - BT$
- Display the process process ID, Arrival Time, Burst Time, Completion Time, Turnaround Time and Waiting Time

Code:

```
#include <iostream>
using namespace std;
int main()
{
    int n;
    cout << "Enter number of processes: ";
    cin >> n;
    int A[n], B[n], C[n], TA[n], WT[n];
    for (int i = 0; i < n; i++)
    {
        cout << "Enter Arrival Time and Burst Time for Process " << i +
1 << ": ";
        cin >> A[i] >> B[i];
    }
    for (int i = 0; i < n; i++)
    {
        for (int j = i + 1; j < n; j++)
        {
            if (B[i] > B[j])
            {
                swap(A[i], A[j]);
                swap(B[i], B[j]);
            }
        }
    }
    C[0] = A[0] + B[0];
    for (int i = 1; i < n; i++)
    {
        if (C[i - 1] < A[i])
            C[i] = A[i] + B[i];
        else
            C[i] = C[i - 1] + B[i];
    }
    for (int i = 0; i < n; i++)
    {
        TA[i] = C[i] - A[i];
        WT[i] = TA[i] - B[i];
    }
    cout << "\nAT\tBT\tCT\tTA\tWT\n";
    for (int i = 0; i < n; i++)
    {
```

```

        cout << A[i] << "\t"
            << B[i] << "\t"
            << C[i] << "\t"
            << TA[i] << "\t"
            << WT[i] << endl;
    }
    return 0;
}

```

Output:

```

"E:\#S-05-CSE-208\Lab Repo" x + v
Enter number of processes: 5
Enter Arrival Time and Burst Time for Process 1: 1 8
Enter Arrival Time and Burst Time for Process 2: 3 7
Enter Arrival Time and Burst Time for Process 3: 2 9
Enter Arrival Time and Burst Time for Process 4: 4 3
Enter Arrival Time and Burst Time for Process 5: 2 3

AT      BT      CT      TA      WT
4        3        7        3        0
2        3       10        8        5
3        7       17       14        7
1        8       25       24       16
2        9       34       32       23

Process returned 0 (0x0)   execution time : 66.374 s
Press any key to continue.

```